

# AWS-Native Scalable Disk Monitoring Solution

## ***Multi-Account, Multi-Region Architecture with AWS Organizations***

*Replacing Ansible while AWS Systems Manager & CloudWatch*

### Executive Summary

As organizations grow through acquisitions and cloud adoption, managing disk utilization across dozens of AWS accounts spanning multiple regions becomes operationally complex. This document presents a fully AWS-native solution that replaces Ansible-based monitoring with AWS Systems Manager (SSM) and Amazon CloudWatch — governed by AWS Organizations for enterprise-scale automation.

The architecture adopts a hub-and-spoke model: a central monitoring account (designated as Delegated Administrator) receives disk metrics from all spoke accounts and regions via CloudWatch Observability Access Manager (OAM). Service-managed CloudFormation StackSets target Organizational Units (OUs) and automatically deploy the complete monitoring stack to any account joining the Organization — zero manual intervention required.

Service Control Policies (SCPs) at the OU level prevent spoke accounts from disabling or removing monitoring components, ensuring compliance is enforced at the governance layer — not just as a recommendation. EventBridge Organization-wide rules route alarm events from all spoke accounts to a central SNS topic for unified alerting

### Problem Statement

| Challenge           | Detail                                                                                                                                                                              |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Account Scale | Enterprise grown through acquisitions: multiple AWS accounts under a single AWS Organization, managed centrally via Management Account and OUs                                      |
| Multi-Region        | Accounts span multiple regions (us-east-1, eu-west-1, ap-south-1, etc.); monitoring infra must be deployed per active region                                                        |
| Ansible Dependency  | Existing Ansible-based monitoring requires SSH access, bastion hosts, key management, and a dedicated Ansible Tower/AWX server — all being removed in favour of AWS-native services |
| Dynamic Instances   | New EC2 instances launch continuously; monitoring enrollment must be automatic with no manual inventory updates                                                                     |
| Governance Gap      | Without centralized policy enforcement, spoke accounts can inadvertently disable monitoring. SCPs close this gap at the Organization level.                                         |

## Solution Architecture

The solution uses an AWS Organizations-governed hub-and-spoke architecture. The Management Account owns the Organization structure and deploys service-managed StackSets. A designated Central Monitoring Account acts as the Delegated Administrator for CloudWatch and SSM, hosting the OAM Sink, CloudWatch Dashboard, SNS topics, and EventBridge rules. Spoke Accounts reside in OUs and receive the complete monitoring stack automatically upon joining the OU.

### Architecture Flow Diagram

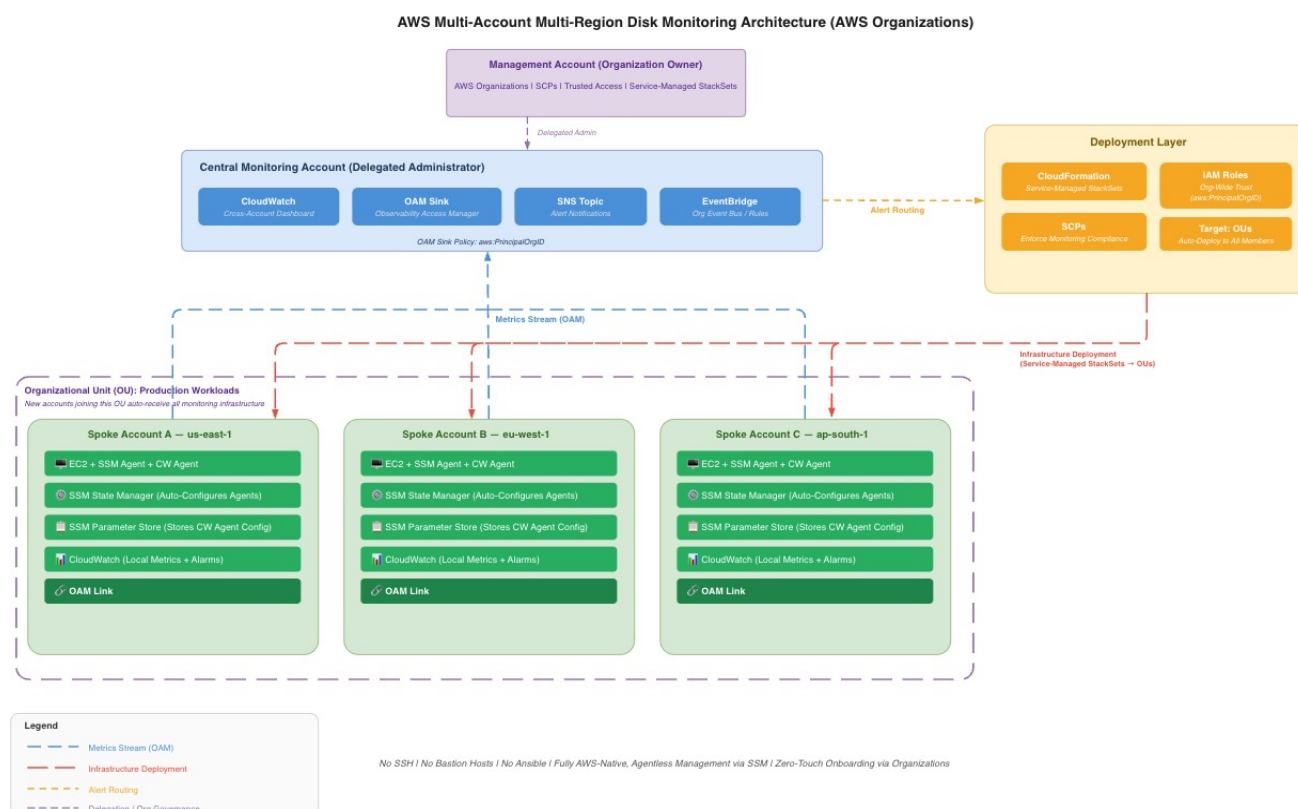


Figure: AWS Organizations Hub-and-Spoke Disk Monitoring Architecture

### Key AWS Services

| Service                          | Role                                                           | Why Chosen                                     |
|----------------------------------|----------------------------------------------------------------|------------------------------------------------|
| <b>AWS Organizations</b>         | Governs all accounts, OUs, SCPs, and service-managed StackSets | Zero-touch auto-deployment to new accounts     |
| <b>Service-Managed StackSets</b> | Deploy monitoring stack to all accounts in an OU automatically | No manual role setup per account required      |
| <b>AWS Systems Manager</b>       | Agent management, remote execution, State Manager associations | Pre-installed on AMIs, no SSH, no bastion      |
| <b>CloudWatch + OAM</b>          | Collect disk metrics; stream from all spokes to central sink   | Native integration, no Lambda pipelines needed |

|                                 |                                                              |                                                           |
|---------------------------------|--------------------------------------------------------------|-----------------------------------------------------------|
| <b>Service Control Policies</b> | Prevent spoke accounts disabling SSM, CW Agent, or OAM Links | Compliance enforcement at governance layer                |
| <b>EventBridge Org Rules</b>    | Route alarm events from all spoke accounts to central SNS    | Unified on-call workflow without per-region subscriptions |
| <b>Delegated Administrator</b>  | Central Monitoring Account manages CloudWatch org-wide       | Management Account stays clean; ops in dedicated account  |

## AWS Organizations Governance Layer

### Trusted Access & Delegated Administrator

Enable trusted access for CloudFormation StackSets and CloudWatch from the Management Account. Designate the Central Monitoring Account as the Delegated Administrator so it can manage cross-account observability without using the Management Account for day-to-day operations.

```
# Run in the MANAGEMENT ACCOUNT
```

```
# Step 1: Enable trusted access for CloudFormation StackSets
```

```
aws organizations enable-aws-service-access \
  --service-principal cloudformation.amazonaws.com
```

```
# Step 2: Enable trusted access for CloudWatch (cross-account observability)
```

```
aws organizations enable-aws-service-access \
  --service-principal cloudwatch.amazonaws.com
```

```
# Step 3: Register Central Monitoring Account as Delegated Administrator
```

```
aws organizations register-delegated-administrator \
  --account-id CENTRAL-MONITORING-ACCOUNT-ID \
  --service-principal cloudformation.amazonaws.com
```

```
aws organizations register-delegated-administrator \
  --account-id CENTRAL-MONITORING-ACCOUNT-ID \
  --service-principal cloudwatch.amazonaws.com
```

### Service Control Policies (SCPs) — Enforce Monitoring Compliance

SCPs are attached to the production OU. They prevent spoke accounts from disabling or removing monitoring infrastructure. This ensures compliance is enforced at the governance layer — even if an account administrator attempts to remove monitoring components.

```
// SCP: Deny-DisableMonitoring — attach to Production OU
```

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyDisableSSMAgent",
      "Effect": "Deny",
      "Action": [
        "ssm:DeregisterManagedInstance",
        "ssm:DeleteAssociation"
      ]
    }
  ]
}
```

```

    ],
    "Resource": "*",
    "Condition": {
      "ArnNotLike": {
        "aws:PrincipalArn": "arn:aws:iam::*:role/StackSet-DiskMonitoring-*"
      }
    }
  },
  {
    "Sid": "DenyDeleteOAMLink",
    "Effect": "Deny",
    "Action": "oam:DeleteLink",
    "Resource": "*",
    "Condition": {
      "ArnNotLike": {
        "aws:PrincipalArn": "arn:aws:iam::*:role/StackSet-DiskMonitoring-*"
      }
    }
  },
  {
    "Sid": "DenyModifyMonitoringRole",
    "Effect": "Deny",
    "Action": [
      "iam:DeleteRole",
      "iam:DetachRolePolicy",
      "iam:DeleteRolePolicy"
    ],
    "Resource": "arn:aws:iam::*:role/DiskMonitoringEC2Role",
    "Condition": {
      "ArnNotLike": {
        "aws:PrincipalArn": "arn:aws:iam::*:role/StackSet-DiskMonitoring-*"
      }
    }
  }
]
}

```

## Component 1: Access & Identity Management

With AWS Organizations, IAM trust no longer requires explicit account IDs. The `aws:PrincipalOrgID` condition key grants trust to any account within the Organization — new accounts are automatically trusted as soon as they join. The SSM Agent communicates via HTTPS outbound (port 443 only), eliminating SSH keys, bastion hosts, and open inbound security group rules.

### Ansible vs. SSM: Access Comparison

| Aspect       | Ansible Approach                                       | SSM Approach (AWS-Native)                               |
|--------------|--------------------------------------------------------|---------------------------------------------------------|
| Connectivity | SSH/WinRM inbound ports open; bastion hosts per region | HTTPS outbound port 443 only; no inbound ports required |

|                       |                                                    |                                                                |
|-----------------------|----------------------------------------------------|----------------------------------------------------------------|
| <b>Authentication</b> | SSH key pairs, Ansible Vault, manual rotation      | IAM instance profiles — auto-rotated, zero secrets to manage   |
| <b>Multi-Account</b>  | Separate inventory and SSH config per account      | aws:PrincipalOrgID — single condition covers all Org accounts  |
| <b>New Account</b>    | Manual key distribution and inventory registration | Automatic — account joins OU and receives all monitoring infra |
| <b>Compliance</b>     | Manual audit of SSH access logs per region         | CloudTrail logs every SSM action; SCPs enforce policy          |

## IAM Trust Policy — Using aws:PrincipalOrgID (Organizations)

The cross-account role uses aws:PrincipalOrgID instead of explicit account IDs. Any account within the Organization can assume this role — no per-account trust policy updates needed when adding new accounts.

```
// IAM Trust Policy — Org-wide cross-account access (no explicit account IDs)
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::CENTRAL-MONITORING-ACCOUNT-ID:root"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "aws:PrincipalOrgID": "o-your-org-id"
        }
      }
    }
  ]
}
```

## EC2 Instance Profile Policy (SSM + CloudWatch)

```
// EC2 Instance Profile Policy — SSM + CloudWatch permissions
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ssm:UpdateInstanceInformation",
        "ssm:ListAssociations",
        "ssm:ListInstanceAssociations",
        "ssmmessages:CreateControlChannel",
        "ssmmessages:CreateDataChannel",
        "ssmmessages:OpenControlChannel",

```

```

    "ssmmessages:OpenDataChannel",
    "ec2messages:GetMessages",
    "ec2messages:AcknowledgeMessage",
    "ec2messages:SendReply"
  ],
  "Resource": "*"
},
{
  "Effect": "Allow",
  "Action": [
    "cloudwatch:PutMetricData",
    "logs:CreateLogGroup",
    "logs:CreateLogStream",
    "logs:PutLogEvents"
  ],
  "Resource": "*"
},
{
  "Effect": "Allow",
  "Action": ["ssm:GetParameter"],
  "Resource": "arn:aws:ssm:*:*:parameter/AmazonCloudWatch-*"
}
]
}

```

## Component 2: Data Collection & Aggregation

The CloudWatch Agent is deployed and managed via SSM. It collects disk metrics every 60 seconds and publishes them to a custom namespace `DiskMonitoring/Custom`. Dimensions include `InstanceId`, `AccountId`, `region`, `mount path`, `device`, and `filesystem type` — enabling granular cross-account, cross-region filtering in dashboards.

### CloudWatch Agent Configuration (JSON)

This configuration is stored centrally in SSM Parameter Store as `AmazonCloudWatch-DiskMonitoring` in each active region. The agent pulls the config on startup and whenever SSM State Manager re-applies the association.

```

// CloudWatch Agent Configuration — cw-agent-config.json
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "root"
  },
  "metrics": {
    "namespace": "DiskMonitoring/Custom",
    "append_dimensions": {
      "InstanceId": "${aws:InstanceId}",
      "AutoScalingGroupName": "${aws:AutoScalingGroupName}"
    },
    "metrics_collected": {

```

```

"disk": {
  "measurement": [
    "used_percent",
    "free",
    "total",
    "inodes_free"
  ],
  "metrics_collection_interval": 60,
  "resources": ["*"],
  "ignore_file_system_types": ["sysfs", "devtmpfs", "tmpfs", "overlay"]
}
}
}
}

```

## Store Config in SSM Parameter Store (CLI)

```

# Bash — Store CloudWatch Agent config in SSM Parameter Store (run per active region)
aws ssm put-parameter \
  --name "AmazonCloudWatch-DiskMonitoring" \
  --type "String" \
  --value file://cw-agent-config.json \
  --description "CloudWatch Agent config for disk utilization monitoring" \
  --tier "Standard" \
  --region <SPOKE_REGION>

```

## Component 3: Scalability — Service-Managed StackSets

Service-managed StackSets are the key differentiator from the previous non-Organization approach. They target Organizational Units (OUs) rather than explicit account IDs. When automatic deployments are enabled, any account joining the OU automatically receives the complete monitoring stack — no manual action, no role creation, no inventory updates.

### Zero-Touch Onboarding

Account joins Production OU → StackSet auto-deploys IAM role, instance profile, SSM Parameter Store config, CloudWatch Agent associations, alarms, SNS topic, and OAM Link — all without any manual intervention. New EC2 instances in that account are automatically enrolled by SSM State Manager (targets InstanceIds: "\*").

## Service-Managed StackSet CloudFormation Template

Deploy this template as a service-managed StackSet from the Management Account (or Delegated Administrator account), targeting the Production OU with automatic deployments enabled. It will deploy to all current member accounts and any future accounts that join the OU.

```

# YAML CloudFormation — Service-Managed StackSet (deploy from Management Account)
AWSTemplateFormatVersion: "2010-09-09"
Description: >
  Disk monitoring infrastructure — deployed via service-managed StackSets
  to all accounts in the Production OU. Auto-deploys to new accounts on OU join.

Resources:
  EC2MonitoringRole:

```

Type: AWS::IAM::Role

Properties:

RoleName: DiskMonitoringEC2Role

AssumeRolePolicyDocument:

Version: "2012-10-17"

Statement:

- Effect: Allow

Principal:

Service: ec2.amazonaws.com

Action: sts:AssumeRole

ManagedPolicyArns:

- arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore

- arn:aws:iam::aws:policy/CloudWatchAgentServerPolicy

EC2InstanceProfile:

Type: AWS::IAM::InstanceProfile

Properties:

InstanceProfileName: DiskMonitoringInstanceProfile

Roles:

- !Ref EC2MonitoringRole

# OAM Link — connects spoke account metrics to the central OAM Sink

OAMLink:

Type: AWS::Oam::Link

Properties:

LabelTemplate: \$AccountName

ResourceTypes:

- AWS::CloudWatch::Metric

- AWS::Logs::LogGroup

SinkIdentifier: !Sub >

arn:aws:oam:\${AWS::Region}:CENTRAL-ACCOUNT-ID:sink/SINK-ID

DiskUtilizationAlarm:

Type: AWS::CloudWatch::Alarm

Properties:

AlarmName: HighDiskUtilization-Warning

AlarmDescription: "Disk utilization exceeds 85% - Warning"

Namespace: DiskMonitoring/Custom

MetricName: disk\_used\_percent

Statistic: Maximum

Period: 300

EvaluationPeriods: 2

Threshold: 85

ComparisonOperator: GreaterThanThreshold

TreatMissingData: breaching

AlarmActions:

- !Ref DiskAlertTopic

DiskUtilizationCritical:

Type: AWS::CloudWatch::Alarm

Properties:

AlarmName: HighDiskUtilization-Critical

```
AlarmDescription: "CRITICAL - Disk utilization exceeds 95%"
```

```
Namespace: DiskMonitoring/Custom
```

```
MetricName: disk_used_percent
```

```
Statistic: Maximum
```

```
Period: 60
```

```
EvaluationPeriods: 1
```

```
Threshold: 95
```

```
ComparisonOperator: GreaterThanThreshold
```

```
TreatMissingData: breaching
```

```
AlarmActions:
```

```
- !Ref DiskAlertTopic
```

```
DiskAlertTopic:
```

```
Type: AWS::SNS::Topic
```

```
Properties:
```

```
TopicName: DiskUtilizationAlerts
```

```
DisplayName: Disk Space Alerts
```

```
DiskAlertSubscription:
```

```
Type: AWS::SNS::Subscription
```

```
Properties:
```

```
TopicArn: !Ref DiskAlertTopic
```

```
Protocol: email
```

```
Endpoint: ops-team@yourcompany.com
```

## SSM State Manager Associations (CloudFormation YAML)

These two associations are deployed into every spoke account and region via the StackSet. The first installs the CloudWatch Agent package; the second applies the disk monitoring configuration from Parameter Store. By targeting InstanceIds: "\*", they catch all current and future EC2 instances automatically.

```
# YAML — SSM State Manager Associations (included in StackSet template)
```

```
Resources:
```

```
InstallCloudWatchAgent:
```

```
Type: AWS::SSM::Association
```

```
Properties:
```

```
AssociationName: DeployCloudWatchAgentForDiskMonitoring
```

```
Name: AWS-ConfigureAWSPackage
```

```
Targets:
```

```
- Key: InstanceIds
```

```
Values:
```

```
- "*"
```

```
Parameters:
```

```
action:
```

```
- Install
```

```
name:
```

```
- AmazonCloudWatchAgent
```

```
ScheduleExpression: "rate(1 day)"
```

```
ComplianceSeverity: HIGH
```

```
ConfigureCloudWatchAgent:
```

```
Type: AWS::SSM::Association
```

```
Properties:
```

AssociationName: ConfigureCloudWatchAgentDiskMetrics  
Name: AmazonCloudWatch-ManagedAgent  
Targets:  
- Key: InstanceIds  
Values:  
- "\*"   
Parameters:  
action:  
- configure  
mode:  
- ec2  
optionalConfigurationSource:  
- ssm  
optionalConfigurationLocation:  
- AmazonCloudWatch-DiskMonitoring  
optionalRestart:  
- "yes"  
ScheduleExpression: "rate(1 day)"  
ComplianceSeverity: HIGH

## Scaling: Organizations vs. Self-Managed

| Growth Scenario                | With AWS Organizations (This Solution)                                                                                      |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| New account added              | Move account into Production OU. StackSet auto-deploys all infra within minutes. Zero manual action.                        |
| New region in existing account | Add region to StackSet deployment targets. SSM associations and CW Agent config redeploy automatically.                     |
| New EC2 instance               | No action needed. SSM State Manager targets InstanceIds: "*" and enrolls new instances on schedule (rate: 1 day).           |
| Config change                  | Update SSM Parameter Store value. State Manager re-applies the new config across all instances in all accounts on next run. |
| Org OU restructure             | Moving account between OUs automatically adjusts SCP application and StackSet targeting based on OU membership.             |

## Component 4: Centralized Dashboard & Observability (OAM)

### What is CloudWatch OAM?

| OAM Component | Where It Lives             | Purpose                                                                                  |
|---------------|----------------------------|------------------------------------------------------------------------------------------|
| Sink          | Central Monitoring Account | Destination that receives all disk metrics from all spoke accounts/regions via OAM Links |

|                    |                                         |                                                                                            |
|--------------------|-----------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Sink Policy</b> | Central Monitoring Account              | Uses aws:PrincipalOrgID — all Org accounts trusted automatically, no explicit account list |
| <b>Link</b>        | Each spoke account x each active region | Connects that account+region CloudWatch data to the central sink. Deployed via StackSets.  |

## OAM Setup with Organization-Wide Policy (CLI)

The sink policy uses aws:PrincipalOrgID instead of explicit account IDs. This means all current and future Organization accounts are automatically permitted to create links — the policy never needs to be updated when accounts are added.

```
#!/bin/bash — Run in the CENTRAL MONITORING account
```

```
# Step 1: Create an OAM Sink in the central monitoring account
```

```
aws oam create-sink \
  --name "DiskMonitoringCentralSink" \
  --tags Key=Purpose,Value=DiskMonitoring
```

```
# Step 2: Create sink policy using aws:PrincipalOrgID (Organization-wide)
```

```
# This allows ANY account in the Org to create OAM Links — no explicit account IDs needed
```

```
aws oam put-sink-policy \
  --sink-identifier "arn:aws:oam:us-east-1:CENTRAL-ACCOUNT-ID:sink/SINK-ID" \
  --policy '{
    "Version": "2012-10-17",
    "Statement": [{
      "Effect": "Allow",
      "Principal": "*",
      "Action": ["oam:CreateLink", "oam:UpdateLink"],
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:PrincipalOrgID": "o-your-org-id"
        }
      },
      "ForAllValues:StringEquals": {
        "oam:ResourceTypes": [
          "AWS::CloudWatch::Metric",
          "AWS::Logs::LogGroup"
        ]
      }
    }]
  }'
```

```
# Step 3: In each spoke account+region, create link (deployed via StackSets — see CFn template)
```

```
aws oam create-link \
  --label-template '$AccountName' \
  --resource-types "AWS::CloudWatch::Metric" \
  --sink-identifier "arn:aws:oam:us-east-1:CENTRAL-ACCOUNT-ID:sink/SINK-ID" \
  --region <SPOKE_REGION>
```

## Dashboard Capabilities

- Disk utilization heatmap across all accounts and regions (filterable by Account, Region, Instance, Mount Point)
- Tag-based filtering using common instance tags shared across all Org accounts
- Anomaly detection widgets for trending disk growth (ML-based, no manual threshold tuning)
- Top-N instances by disk utilization ranked across all accounts and all regions
- Alarm state overview — see all WARNING/CRITICAL alarms at a glance from the central account

## Component 5: Alerting & EventBridge Organization Rules

CloudWatch Alarms are deployed to every spoke account and region via StackSets. EventBridge Organization-wide rules route alarm events from all spoke accounts to the central SNS topic — enabling a unified on-call workflow without requiring per-account or per-region subscriptions.

| Alarm        | Threshold | Evaluation        | Trigger Condition                                 | Action                         |
|--------------|-----------|-------------------|---------------------------------------------------|--------------------------------|
| Warning      | > 85%     | 2 x 5-min periods | Disk filling — cleanup needed in 24-48h           | SNS email to ops-team          |
| Critical     | > 95%     | 1 x 1-min period  | Imminent disk full — immediate action required    | SNS + optional PagerDuty/Slack |
| Missing Data | No data   | Immediate         | Agent stopped reporting — instance or agent issue | SNS alert (breaching state)    |

### EventBridge Organization Rule (Cross-Account Alert Routing)

An EventBridge Organization rule deployed from the Management Account routes CloudWatch Alarm state changes from ALL spoke accounts to the central SNS topic. No per-account EventBridge configuration needed.

```
// EventBridge Organization Rule — deploy from Management Account
{
  "Name": "DiskMonitoringOrgAlarmRoute",
  "EventPattern": {
    "source": ["aws.cloudwatch"],
    "detail-type": ["CloudWatch Alarm State Change"],
    "detail": {
      "alarmName": [{"prefix": "HighDiskUtilization"}],
      "state": {
        "value": ["ALARM"]
      }
    }
  },
  "Targets": [
    {
      "Id": "CentralSNSTopic",
      "Arn": "arn:aws:sns:us-east-1:CENTRAL-ACCOUNT-ID:DiskUtilizationAlerts",
      "RoleArn": "arn:aws:iam::CENTRAL-ACCOUNT-ID:role/EventBridgeOrgAlertRole"
    }
  ]
}
```

```

}
]
}

# CLI: Create Organization-wide event bus rule
aws events put-rule \
  --name "DiskMonitoringOrgAlarmRoute" \
  --event-pattern file://alarm-pattern.json \
  --state ENABLED \
  --event-bus-name "arn:aws:events:us-east-1:MANAGEMENT-ACCOUNT-ID:event-bus/default"

# Enable EventBridge cross-account resource policy on central bus
aws events put-permission \
  --action events:PutEvents \
  --principal "*" \
  --condition '{"Type":"StringEquals","Key":"aws:PrincipalOrgID","Value":"o-your-org-id"}' \
  --statement-id AllowOrgAccounts

```

## Advantages of AWS-Native Services Over Ansible

Replacing Ansible with AWS Systems Manager, CloudWatch, and AWS Organizations eliminates multiple layers of operational overhead while delivering superior security, scalability, and compliance enforcement.

| Area                     | Ansible                                                     | AWS-Native (Org + SSM + CloudWatch)                                    |
|--------------------------|-------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Connectivity</b>      | SSH/WinRM inbound ports; bastion hosts per region; open SGs | HTTPS outbound port 443 only; no inbound ports; no bastion             |
| <b>Credentials</b>       | SSH key pairs, Ansible Vault, manual rotation per account   | IAM roles with Org-wide trust — auto-rotated, zero secrets             |
| <b>Inventory</b>         | Static host files or dynamic scripts per account/region     | Auto-discovery via SSM Fleet Manager (per region, real-time)           |
| <b>New Account</b>       | Manual inventory, SSH key distribution per new account      | Join OU → zero-touch auto-deploy via service-managed StackSets         |
| <b>Multi-Region</b>      | Separate inventory per region; complex jump host routing    | Regional SSM + service-managed StackSets deploy natively               |
| <b>High Availability</b> | Dedicated Ansible Tower/AWX EC2 — HA = complex & costly     | Fully managed, serverless — AWS manages availability per region        |
| <b>Drift Detection</b>   | Manual playbook re-runs on a schedule; no feedback loop     | State Manager continuously enforces desired state & reports compliance |
| <b>Compliance</b>        | No guardrails — admins can remove monitoring at will        | SCPs prevent disabling monitoring at governance layer                  |

|                         |                                                           |                                                                       |
|-------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Security Posture</b> | Large attack surface: SSH, bastion, keys per region       | Minimal attack surface: IAM + HTTPS + SCP guardrails                  |
| <b>Audit Trail</b>      | Manual aggregation of SSH logs across accounts/regions    | CloudTrail logs every SSM action per region with IAM identity         |
| <b>Visibility</b>       | Custom aggregation stack (Elasticsearch/Grafana) required | Native OAM — metrics flow to central account; zero config             |
| <b>Cost</b>             | Tower license + dedicated EC2 + operational overhead      | SSM free; CW Agent free; minimal metric/alarm charges (~\$200-300/mo) |

## Key Differentiators:

1. **Zero-Touch Onboarding with Organizations:** New accounts receive full monitoring infrastructure simply by joining the Production OU. StackSets deploy within minutes automatically — no IAM role creation, no inventory updates, no playbook runs.
2. **SCP-Enforced Compliance:** Service Control Policies at the OU level prevent spoke accounts from disabling SSM, removing OAM Links, or deleting monitoring IAM roles — even with Administrator access. Compliance is governance-enforced.
3. **No SSH, No Bastion Hosts:** SSM uses HTTPS outbound via the AWS backbone. Eliminates key rotation, bastion host costs, and open inbound security group rules across every account and region.
4. **No Dedicated Servers:** Ansible Tower/AWX requires a running EC2 with HA configuration per region. SSM is fully managed — no infrastructure to operate, patch, or monitor.
5. **Self-Healing by Design:** SSM State Manager re-applies desired configuration if drift occurs in any account or region, on schedule, automatically. The SCPs prevent drift from being permanent.
6. **Organization-Wide OAM Policy:** The sink policy uses aws:PrincipalOrgID. All Org accounts can link without any policy updates when accounts are added — truly zero-config observability scaling.
7. **Delegated Administrator Pattern:** The Management Account stays clean (no operational workloads). The Central Monitoring Account is the Delegated Administrator, enabling cross-account CloudWatch management without Management Account credentials.
8. **EventBridge Organization Rules:** A single EventBridge rule in the Management Account routes alarm events from ALL member accounts to the central SNS topic — no per-account or per-region configuration.

## Cost Analysis

The AWS-native Organizations solution is significantly more cost-efficient than Ansible Tower/AWX. Most core components are either free or have negligible per-unit pricing. Cost scales linearly with the number of instances — NOT with the number of accounts or regions added.

| Component        | Monthly Cost | Notes                                                            |
|------------------|--------------|------------------------------------------------------------------|
| <b>SSM Agent</b> | <b>Free</b>  | Pre-installed on Amazon Linux 2, Amazon Linux 2023, Windows AMIs |

|                                  |                      |                                                                   |
|----------------------------------|----------------------|-------------------------------------------------------------------|
| <b>SSM State Manager</b>         | <b>Free</b>          | Free for AWS-managed documents (AWS-ConfigureAWSPackage, etc.)    |
| <b>CloudWatch Agent</b>          | <b>Free</b>          | Software only; metric publication charges apply separately        |
| <b>AWS Organizations</b>         | <b>Free</b>          | No charge for Organizations, SCPs, OUs, or StackSets permissions  |
| <b>CloudFormation StackSets</b>  | <b>Free</b>          | No additional charge for service-managed StackSet operations      |
| <b>CloudWatch Custom Metrics</b> | ~\$0.30/metric       | ~4 metrics/disk x instances x mount points (first 10K metrics)    |
| <b>CloudWatch Alarms</b>         | \$0.10/alarm         | 2 alarms per instance (Warning 85% + Critical 95%)                |
| <b>CloudWatch OAM</b>            | <b>Free</b>          | No additional charge for metric sharing via OAM Links             |
| <b>SNS Notifications</b>         | <b>First 1M free</b> | Email deliveries; HTTP/S \$0.60/million; very low volume          |
| <b>EventBridge (Org rules)</b>   | \$1.00/M events      | For cross-account alarm routing; alarm events are very low volume |

## Conclusion

This AWS Organizations-based disk monitoring solution provides the most streamlined, secure, and scalable approach for multi-account, multi-region environments. By combining Organizations governance (SCPs, OUs, service-managed StackSets), SSM (agentless management), CloudWatch OAM (unified observability), and EventBridge Organization rules (centralized alerting), the architecture eliminates Ansible entirely while delivering superior automation.

New accounts receive full monitoring infrastructure — IAM roles, CloudWatch Agent, alarms, SNS topics, and OAM Links — simply by joining the Production OU. No manual intervention, no role creation, no inventory updates. SCPs ensure that monitoring cannot be accidentally or intentionally disabled by spoke account administrators, enforcing compliance at the governance layer.

The centralized CloudWatch dashboard with Organization-wide OAM provides true single-pane-of-glass visibility, with cross-account and cross-region metric queries available natively. EventBridge Organization rules deliver a unified on-call experience without per-account configuration. The total cost for monitoring 500 VMs across 5 accounts and 3 regions is approximately \$200-300/month — a fraction of the Ansible Tower licensing and infrastructure cost it replaces.